

CHE253K, APPLIED STATISTICS
UNIQUE NUMBER: 17070

Fall 2026

MEETING TIME AND LOCATION: 9:00 – 10:00 am, Monday, Wednesday, and Friday, CPE 2.218

INSTRUCTOR

Dr. William Howard, Associate Professor of Practice
Please address as Professor Howard or Dr. Howard
Office*: CPE 5.462

Email: whoward@che.utexas.edu

Please conduct all communication through Canvas. Use the above email address only as a backup.

Office hours: Wednesdays, 10:00 – 12:00, CPE 5.462, and by appointment

*Carrying Handguns in my Personal Office

Per my right, I prohibit carrying of handguns in my personal office. Note that this information will also be conveyed to all students verbally during the first week of class. This written notice is intended to reinforce the verbal notification and is not a “legally effective” means of notification in its own right.”

TEACHING ASSISTANT

Jiyong Lee, Graduate Student, UT Chemical Engineering
Please address as Jiyong

Email: jiyonglee@utexas.edu

Office hours and location: Tuesday: 1:00–2:00 PM, Friday: 1:00–2:00 PM, Location: CPE 3.448

TUTOR

Trinisha Nguyen, Undergraduate Student, UT Chemical Engineering
Please address as Trinisha

Email: trinisha@utexas.edu

Office hours and location: Monday 12:30-3:00 PM and Tuesday 11:00 AM-12:00 PM, CHE East Lab (CPE 2.702)

FINAL EXAM

Currently scheduled for: Monday, December 14, 10:30 AM-12:30 PM. Location TBD.

Check at this URL: <https://onestop.utexas.edu/registration-and-degree-planning/final-exam-information/>

COURSE DESCRIPTION

UNIVERSITY CATALOG COURSE DESCRIPTION

Statistical methods such as data exploration and summary, least-squares fitting, probability and probability distributions, statistical inference and hypothesis testing, analysis of variance, design of experiments, statistical quality control, and use of professional statistical software. Chemical engineering majors must make a grade of at least C- in this course.

PRE-REQUISITES FOR THE COURSE

Credit with a grade of at least a C- or registration for Chemical Engineering 317; and the following coursework with a grade of at least a C- in each Chemical Engineering 210 and Mathematics 427J.

LEARNING OUTCOMES

LO1: Define and apply the basic rules of probability and probability distributions.

LO2: Formulate and test statistical hypotheses.

LO3: Generate linear and nonlinear regression models from data.

LO4: Design an experimental plan.

LO5: Deploy these skills in software such as Excel and JMP

HOW WILL YOU LEARN?

COURSE CANVAS SITE: <https://utexas.instructure.com/courses/1458951>

COMMUNICATION

Please email me through Canvas. You are responsible for ensuring that the primary email address you have recorded with the university is the one you will check for course communications because that is the email address that Canvas uses.

TEACHING MODALITY INFORMATION

1. The course will be taught “face-to-face” in the classroom, as noted in the University Catalog Course Description. Per guidance from the Chair of the Department, the course format might be adjusted in response to a significant event such as a surge in COVID cases. Students should plan to meet face-to-face for the full semester.
2. The textbook is ideally suited to the learning objectives of the class. For most of the course, we will follow the textbook in order. There will be some exceptions.
3. Most classes will be a combination of lecture, working problems together as a class or in small groups, watching short videos that illustrate a process or key idea, and taking a daily graded quiz that will assess and reinforce learning.
4. The daily graded quizzes are designed to assess student learning of the most important concepts in the assigned reading and the lecture.

Other than normal emergency accommodations, there will be no alternative to in-person attendance.

ASKING FOR HELP

Time will be allocated for questions during the lectures. Also, students may attend office hours, seek help from the TA(s) and/or tutor(s), or request help via an appointment.

ACADEMIC ACCOMMODATIONS

1. Students with disabilities may request appropriate academic accommodations from the Division of Student Affairs, Disability and Access (D&A) – see the [University Policies and Resources for Students](#) Canvas page or visit the [D&A website](#).
2. Students with accommodations should talk with me (Professor Howard) as soon as this is possible to be sure I have received the email notification and to discuss your accommodations.

COURSE REQUIREMENTS AND GRADING

REQUIRED MATERIALS

1. “Probability and Statistics for Engineering and the Sciences, 9th ed.”, Jay L. Devore, Cengage Learning, Boston, MA. (ISBN: 9781305251809)
2. The textbook for this course can be found at this link: <https://www.universitycoop.com/pages/adoption-search-results?courseId%5B%5D=547946>
3. The course adheres to the LTA program.
 - a. Details of this program are provided at the following URL: <https://provost.utexas.edu/initiatives/longhorn-textbook-access/>
 - b. See especially the section at the above site “Frequently Asked Questions LTA” and the question “How can a student opt out of the LTA program?”

SHARING OF COURSE MATERIALS IS PROHIBITED

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without my explicit, my written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

REQUIRED DEVICES

1. Students are required to have an electronic means to access Canvas during class. Daily quizzes will be administered via Canvas.
2. Students are required to have a personal laptop with JMP and Microsoft Excel installed and working. This will be confirmed the first week of class.
3. Students are required to have a separate electronic calculator. This must be a single line calculator that does not have graphing or programming features. You may not use the calculator on your phone.

CONFIDENTIALITY OF CLASS RECORDINGS

1. HOP 2-9970 prohibits students from recording class instruction (audio or video) unless a student obtains the instructor's permission or Disability & Access has approved audio recording as an accommodation.
2. The lecturer plans to make audio recordings of the class sometimes, but not necessarily every time.
3. Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

GETTING HELP WITH TECHNOLOGY

To learn more about available technologies for students or if you require technical assistance, please visit the Technology for Students website.

CLASSROOM EXPECTATIONS

1. Class attendance is not strictly mandatory. However, quizzes occur at the start of almost every class and these account for a significant fraction of the course grade. Quizzes cannot be made up.
2. Interactions in the classroom are a vital part of the learning experience. Students are encouraged to participate in class. Your participation enhances your own learning and benefits the other students in the class. Class participation is not part of the course grade.

BEHAVIOR EXPECTATIONS

1. Students should approach the class eager to learn and participate in the learning process.
2. Students should be prepared for class by completing assigned reading and homework and reviewing past material as needed.
3. There are some things that distract from the learning process:
 - a. Except when necessary for participation in questions and answers and any group discussions or activities, students are expected to be quiet and refrain from any conversation during class. Conversations include audible talking, passing notes, and any form of digital communication.
 - b. Except when using the Canvas application to access a quiz, use of mobile phones during class is prohibited and mobile devices should be put away and not visible to the instructor during class. It is acceptable if mobile devices are with the student, but they should be out of sight.
 - c. Mobile devices and any other electronic devices should be placed in silent mode.
 - d. Use of tablets or laptops for the purposes of taking notes is permitted. However, such use should not interfere with the learning process of other students.

PROFESSIONAL STANDARDS

Any graded work should be legible and clear. Multiple choice questions must have a clearly marked answer that is obvious. Points on graded work will be deducted if the answer is not clear and/or is not presented in a clear way.

CONTENT WARNING

Our classroom provides an open space for the critical and orderly exchange of ideas through discussion. Some readings and other content in this course will include topics and comments that some students may find offensive and/or traumatizing. I'll aim to forewarn students about potentially disturbing content, and I ask all students to help to create an atmosphere of mutual respect and sensitivity.

ARTIFICIAL INTELLIGENCE POLICY STATEMENT

In accordance with the University's Institutional Rules on Student Services and Activities, Chapter 11, students accept the responsibility to always uphold academic integrity and an honor code reflective of a scholarly community devoted to academic and personal success. All members of the University community are fully accountable and responsible for any output they produce as part of academic work. They are also responsible for following the guidance specified in the [Texas Statement on Academic Integrity](#) and avoiding prohibited uses of generative AI tools outlined in the Information Security Office's guidance on [Acceptable Use of Generative AI Tools](#).

In this course, generative [AI use is partially permitted](#) as explained in the following four numbered sections.

1. **DAILY QUIZZES AND ASSIGNMENTS:** Unless it is explicitly stated in writing as part of the quiz or assignment AI of any kind is not allowed during any Quizzes or Assignments.
2. **TESTS AND EXAMS:** AI of any kind is not allowed during any Tests or the Final exam.
3. **INSTALLING, TESTING, LEARNING JMP AND MICROSOFT EXCEL:** AI use is allowed and encouraged to help you install, test, and learn the software applications JMP and Microsoft Excel.
4. Violating these provisions in this course may constitute academic misconduct under UT Austin's Institutional Rules and may be referred to Student Conduct and Academic Integrity in the Office of the Dean of Students for resolution.

ACADEMIC INTEGRITY EXPECTATIONS

All allegations of academic misconduct will be referred by faculty to the Office of the Dean of Students. Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can include a written warning, probation, deferred suspension, suspension, or expulsion from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at <https://deanofstudents.utexas.edu/conduct/student-conduct-processes.php>

ASSIGNMENTS

The following table represents how you will demonstrate your learning and how we will assess the degree to which you have done so.

Assignments	Assessment Method	Percent of Total Grade
Quizzes occur during almost every class.	Quiz grades are converted to percents and the lowest 4 quiz grades for each student will be dropped. Your final score for this component will be the average of your percent scores on the quizzes so that each quiz counts equally.	20%
Homework assignments	Each homework assignment is converted to a percentage grade. Your final score for this component will be the average of your percent scores on the homework assignments so that each homework assignment counts equally.	20%
Test 1 scheduled for 25 September, in our normal classroom	Scored on a percent basis with a maximum of 100%	20%
Test 2 scheduled for 30 October, in our normal classroom	Scored on a percent basis with a maximum of 100%	20%
Final exam scheduled for Monday, December 14, 10:30 AM-12:30 PM. Location TBD.	Scored on a percent basis with a maximum of 100%	20%

RELIGIOUS HOLY DAYS

By UT Austin policy, you must notify me of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence. For questions regarding religious accommodations, please contact the Office of the Dean of Students.

GRADING POLICY

MEANS OF EVALUATION AND ASSIGNMENT OF CLASS GRADES

See table above. Each of the 5 components below represents 20% of the course grade.

See the section titled “Religious Holy Days” which supersedes any other rules provided in this syllabus that concern grading or missing class.

1. Quiz grades are converted to percents and the lowest 4 quiz grades for each student will be dropped. Your final score for this component will be the average of your percent scores on the quizzes so that each quiz counts equally.
2. Homework assignments are converted to a percentage grade. Your final score for this component will be the average of your percent scores on the homework assignments so that each homework assignment counts equally.
3. Test 1: Scored on a percent basis with a maximum of 100%
4. Test 2: Scored on a percent basis with a maximum of 100%
5. Final Exam: Scored on a percent basis with a maximum of 100%

LATE WORK AND MAKING UP MISSED WORK

1. See the section titled “Religious Holy Days” which supersedes any other rules provided in this syllabus that concern grading or missing class.
2. Students must provide a certified medical excuse letter or a letter on university letterhead (for school-related activities) to reschedule any assignment or change/extend a deadline.
3. Late homework is not accepted. The due dates/times are strictly enforced, subject to the exceptions listed in #1 and #2 above.
4. Quizzes cannot be made up. There is a generous allowance that will automatically drop the 4 lowest quiz grades for each student. Students are advised that this policy is not a mechanism to justify missing class but rather to account for cases where students must miss class. Examples include: attend a conference, attend a legal proceeding, attend to an important family matter, attend an interview, attend a funeral, sickness, etc. Excused absences listed below in a separate paragraph do NOT count against this allowance of 4 dropped quizzes.
5. If a student misses Test 1, Test 2, or the Final Exam due to an excused absence, or an exceptional, unforeseen, and unavoidable situation, the student will be required to take a makeup exam. If you missed Test 1, Test 2, or the Final Exam for any other reasons, you will not be allowed to make up the test/exam.

REGRADE REQUESTS

This applies only to homework and tests. Please submit regrade requests **BOTH** as an **EMAIL IN CANVAS** and **IN WRITING** directly to me. Please write your request in the email and on a separate sheet of paper and attach it to the original graded assignment; do not make any changes or annotations to the original document. Regrade requests must be submitted within one calendar week (7 calendar days) of receiving your graded assignment. The day the graded assignment was received counts as day 1.

ABSENCES

Class attendance is not strictly mandatory. However, quizzes occur at the start of almost every class and these account for a significant fraction of the course grade. Quizzes cannot be made up. See also the section “Late Work and making up missed work”

EXCUSED ABSENCES

The following are considered Excused Absences. Students will be excused from the daily quiz, if we have one, on that date.

1. Any excused absences that adhere to the “Religious Holy Days” section of the syllabus.
2. Any excused absences that are documented as related to the individual student’s accommodations, if applicable.
3. An absence in which the student has provided a certified medical excuse letter or a letter on university letterhead (for school-related activities) – and for which the documentation was provided no later than 1 week after the absence.

EXTRA CREDIT

No extra credit opportunities are currently planned for the course. However, the instructor might add these.

GRADE BREAKS

Plus/minus grading will be used.

Grade	Cutoff
A	94%
A-	90%
B+	87%
B	84%
B-	80%
C+	77%
C	74%
C-	70%
D+	67%
D	64%
D-	60%
F	<60%

COURSE OUTLINE

All instructions, assignments, readings, rubrics and essential information will be on the Canvas website at utexas.instructure.com. Check Canvas regularly. Changes to the schedule may be made at my discretion if circumstances require. I will announce any such changes in class and will also communicate them via a Canvas announcement. It is your responsibility to note these changes when announced, and I will do my best to ensure that you are notified of changes with as much advance notice as possible.

FINAL EXAM DATE AND TIME

Currently scheduled for: Monday, December 14, 10:30 AM-12:30 PM. Location TBD.

Check at this URL: <https://onestop.utexas.edu/registration-and-degree-planning/final-exam-information/>

TENTATIVE SCHEDULE

1. The table below provides the tentative schedule for the course.
2. Students should expect that changes will be needed for any number of reasons including an assessment by the instructor that more time is needed on a particular topic.
3. The dates of Test 1, Test 2, and the Final Exam will not change except for truly exceptional and unusual circumstances.

Week	Class #	Day	Date	Class Topic	Graded Activities
1	1	Monday	August 24	Welcome and course introduction	
1	2	Wednesday	August 26	Descriptive statistics part 1	Survey
1	3	Friday	August 28	JMP and Excel skills	Quiz
2	4	Monday	August 31	Descriptive statistics part 2	Quiz
2	5	Wednesday	September 2	Basic concepts of probability	Quiz
2		Friday	September 4	No Recitation – students should use time to complete JMP and Excel assignments	
3		Monday	September 7	NO CLASS, LABOR DAY	
3	6	Wednesday	September 9	Solving probability problems	Quiz
3	7	Friday	September 11	Recitation	Quiz
4	8	Monday	September 14	Conditional probabilities and Bayes Theorem. Independence.	Quiz
4	9	Wednesday	September 16	Discrete random variables and probability distributions	Quiz
4	10	Friday	September 18	Recitation	Quiz
5	11	Monday	September 21	Binomial distribution function and applications	Quiz
5	12	Wednesday	September 23	Review for Test 1	Quiz
5	13	Friday	September 25	Test 1	Test 1
6	14	Monday	September 28	Continuous Random Variables and Probability Distributions.	Quiz
6	15	Wednesday	September 30	Continuous Random Variables and Probability Distributions.	Quiz
6	16	Friday	October 2	Recitation	Quiz
7	17	Monday	October 5	Joint probability distributions and random samples. Central limit theorem	Quiz
7	18	Wednesday	October 7	Joint probability distributions and random samples. Central limit theorem	Quiz
7	19	Friday	October 9	Recitation	Quiz
8	20	Monday	October 12	Confidence intervals	Quiz
8	21	Wednesday	October 14	Confidence intervals	Quiz
8	22	Friday	October 16	Recitation	Quiz
9	23	Monday	October 19	Hypothesis Testing	Quiz
9	24	Wednesday	October 21	Hypothesis Testing	Quiz
9	25	Friday	October 23	Recitation	Quiz
10	26	Monday	October 26	Hypothesis Testing	Quiz
10	27	Wednesday	October 28	Review for Test 2	Quiz
10	28	Friday	October 30	Test 2	Test 2
11	29	Monday	November 2	Flexible topic, possibly ANOVA or case studies of statistics use	Quiz

Week	Class #	Day	Date	Class Topic	Graded Activities
11	30	Wednesday	November 4	Flexible topic, possibly ANOVA or case studies of statistics use	Quiz
11	31	Friday	November 6	Recitation	Quiz
12	32	Monday	November 9	Regression	Quiz
12	33	Wednesday	November 11	Regression	Quiz
12	34	Friday	November 13	Recitation	Quiz
13	35	Monday	November 16	Statistical Process Control	Quiz
13	36	Wednesday	November 18	Statistical Process Control	Quiz
13	37	Friday	November 20	Recitation	Quiz
14		Monday	November 23	Fall break – no class	
14		Wednesday	November 25	Fall break – no class	
14		Friday	November 27	Fall break – no class	
15	38	Monday	November 30	DOE/Response Surface Models	Quiz
15	39	Wednesday	December 2	DOE/Response Surface Models	Quiz
15	40	Friday	December 4	Recitation	Quiz
16	41	Monday		Course review practice problems	Quiz
16		Wednesday		UT STUDY DAY - NO CLASS	
		Monday,	December 14, 10:30 AM- 12:30 PM.	Final Exam Location TBD	Final

UNIVERSITY POLICIES AND RESOURCES

For a list of important university policies and helpful resources that you may need as you engage with and navigate your courses and the university, see the [University Policies and Resources for Students](#) Canvas page. The page includes important safety information including the policy on carrying of handguns on campus, the language of the University Honor Code, Title IX legal requirements for Texas employees, and information about how to receive support through the office of Disability & Access, tutoring services, the Counseling and Mental Health Center, University Health Services, and other resources.

** End of syllabus